

GRPO vs Our Method in verl-gppo

Scope. This note compares:

- **GRPO baseline:** `algorithm.adv_estimator=grpo`.
- **Our method:** `algorithm.adv_estimator=gppo` with `predict_advantage_head=true`.

The equations below are matched to the current code in `verl/trainer/ppo/core_algos.py`, `verl/trainer/ppo/ray_trainer.py`, and `verl/workers/actor/dp_actor.py`.

Shared setup. For each sampled response i , let g_i be its prompt id (`uid`), $m_{i,t} \in \{0,1\}$ be response mask, and token rewards be $r_{i,t}$. Define outcome reward

$$R_i = \sum_t r_{i,t}.$$

Rollouts are grouped by prompt id: $S_g = \{i : g_i = g\}$.

Comparison convention (important). To make the ablation comparable in your intended way:

- **GRPO** uses `loss_agg_mode=seq-mean-token-mean` (classic per-sequence normalization).
- **Our method** uses `loss_agg_mode=seq-mean-token-sum`: token losses are summed within each sequence (no divide by token count), then averaged across sequences.

Let $\ell_{i,t}$ be token-level PPO loss terms and $T_i = \sum_t m_{i,t}$. Then

$$\mathcal{L}_{\text{GRPO}} = \frac{1}{N_{\text{valid}}} \sum_{i: T_i > 0} \frac{1}{T_i} \sum_t m_{i,t} \ell_{i,t},$$

$$\mathcal{L}_{\text{ours}} = \frac{1}{N_{\text{valid}}} \sum_{i: T_i > 0} \sum_t m_{i,t} \ell_{i,t}.$$

Our method additionally replaces grouped sequence-level policy advantages with token-level predicted advantages from the actor head.

1. GRPO (baseline)

For each group g , compute group baseline statistics:

$$\mu_g, \sigma_g \quad \text{from } \{R_i\}_{i \in S_g}.$$

If `use_ipw=true`, weights are proportional to $1/\pi_i$, giving weighted μ_g, σ_g . Then scalar advantage is

$$a_i^{\text{grpo}} = \begin{cases} \frac{R_i - \mu_{g_i}}{\sigma_{g_i} + \varepsilon}, & \text{if std-normalization on,} \\ R_i - \mu_{g_i}, & \text{otherwise.} \end{cases}$$

Token-level fields written to batch:

$$A_{i,t}^{\text{grpo}} = a_i^{\text{grpo}} m_{i,t}, \quad \text{Ret}_{i,t}^{\text{grpo}} = A_{i,t}^{\text{grpo}}.$$

So in this implementation, `returns` are the same tensor as normalized/centered GRPO advantages.

2. Our Method (GPPO + advantage head)

2.1 Grouped target (GPPO path). GPPO also forms grouped scalar targets by prompt id:

$$a_i^{\text{gppe}} = \begin{cases} \frac{R_i - \mu_{g_i}}{\sigma_{g_i} + \varepsilon}, & \text{if std-normalization on,} \\ R_i - \mu_{g_i}, & \text{otherwise.} \end{cases}$$

Then

$$A_{i,t}^{\text{gppe}} = a_i^{\text{gppe}} m_{i,t}, \quad \text{Ret}_{i,t}^{\text{gppe}} = R_i m_{i,t},$$

and an extra scalar field is stored:

$$\text{advantage_targets}_i = a_i^{\text{gppe}}.$$

2.2 Next-token aligned advantage head. Let $H_{i,t}$ be backbone hidden states. Policy and advantage use different hidden representations:

$$h_{i,t}^\pi = H_{i,t}, \quad h_{i,t}^A = W_s H_{i,t} + b_s.$$

Advantage branch predicts action-conditioned scores over vocabulary actions $a \in \{1, \dots, V\}$:

$$u_{i,t,a} = W_A[a]^\top h_{i,t}^A + b_A[a].$$

For the sampled response token $y_{i,t}$, the token advantage prediction is

$$\tilde{z}_{i,t} = u_{i,t-1,y_{i,t}} m_{i,t}, \quad t = 1, \dots, T,$$

which is next-token aligned (state $t-1$ for action token t). Sequence prediction:

$$\hat{a}_i = \sum_{t=1}^T \tilde{z}_{i,t}.$$

Auxiliary regression loss:

$$\mathcal{L}_{\text{adv}} = \frac{1}{B} \sum_{i=1}^B (\hat{a}_i - \text{advantage_targets}_i)^2.$$

Applied as `advantage_loss_coef` · $\mathcal{L}_{\text{adv}}$.

2.3 PPO update in actor. When `predict_advantage_head=true`, policy loss uses detached predicted token advantages:

$$A_{i,t}^{\pi} = \text{stopgrad}(\tilde{z}_{i,t}).$$

Otherwise it uses $A_{i,t}^{\text{gppo}}$.

2.4 Update order. Per training step in this codepath:

1. Compute rewards and grouped advantages.
2. Update critic (if enabled).
3. Update actor; if advantage head is enabled, actor first runs $\mathcal{L}_{\text{adv}}$, then PPO policy loss.

3. Side-by-side summary

Item	GRPO baseline	Our method
Grouping key	prompt id (uid)	prompt id (uid)
Scalar group target	centered (or std-normalized) R_i	same grouped target
advantages	$a_i m_{i,t}$	$a_i m_{i,t}$
returns	same as advantages	broadcast raw R_i
Extra target field	none	<code>advantage_targets = a_i</code>
Extra actor loss	none	MSE on summed next-token-aligned action-conditioned scores
Policy advantage source	grouped target directly	detached head prediction (if enabled)
PPO loss aggregation	<code>seq-mean-token-mean</code>	<code>seq-mean-token-sum</code>
IPW baseline option	yes (<code>use_ipw</code>)	no
Default critic need	no	yes in current <code>need_critic</code> rule

4. Minimal ablation matrix

Run	adv estimator	adv head	critic	loss agg mode
GRPO baseline	<code>grpo</code>	false	false	<code>seq-mean-token-mean</code>
Our method	<code>gppo</code>	true	true	<code>seq-mean-token-sum</code>